

Khandoker Md. Mashiur Rahman

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Objectives

Python and Machine Learning Engineer experienced in developing production-ready AI systems across computer vision, document intelligence, and LLM-based applications. Delivered government and commercial solutions with measurable performance improvements through efficient backend and ML pipeline design.

Experience

Software Engineer

June 2023 – Present

Simec Systems Ltd., Dhaka, Bangladesh

- Served as the primary Python engineer, owning end-to-end development, debugging, and optimization of backend and ML-driven systems.
- Delivered 5+ production projects, including 2 government systems, translating requirements into secure and scalable Python solutions.
- Led computer vision initiatives using OpenCV, achieving reliable image detection and matching performance across real-world datasets.
- Designed and deployed various AI agent platforms using LangChain, LLaMA 3, ChromaDB etc. and improved backend performance by optimizing APIs and data pipelines, reducing response time by up to 80% under high-traffic conditions.

Research and Projects

AI-Powered PDF Q&A Assistant (RAG + OCR)

2025

Project Link: [GitHub](#) [🔗](#)

- Designed a multimodal RAG pipeline using LangChain, LLaMA 3, and ChromaDB to query structured and unstructured PDFs.
- Implemented OCR fallback and table reasoning, improving robustness on scanned documents.
- Enabled multi-turn conversational retrieval with memory and real-time Streamlit interface.

Depth-Based 2D to 3D Point Cloud Generation

2025

Project Link: [GitHub](#) [🔗](#)

- Developed a computer vision pipeline converting 2D RGB images into dense 3D point clouds using MiDaS depth estimation.
- Applied camera projection geometry and transformation matrices to simulate multi-view reconstruction for improved spatial accuracy.
- Generated resolution-agnostic PLY outputs with RGB encoding for visualization in MeshLab and CloudCompare.

DGDP (Director General of Defense Purchase) – Government Project - Backend

2025

Project Link: [Live Project](#) [🔗](#)

- Developed a secure document verification system using Python and Java, integrating encryption and peer-to-peer communication for government-grade data protection.
- Implemented hardware-based authentication via security dongle, achieving robust user verification and reducing document tampering risk by 90%.

- Owned end-to-end development, from system design to deployment, ensuring compliance with security protocols.

LIMS (Labour Information and Management Systems) – Government Project – Computer Vision Module 2024-2025

Project Link: [Live Project](#) 

- Designed and tested an image detection and matching pipeline using OpenCV, achieving 85% identification accuracy on labor records.
- Ensured system robustness across multiple image formats and lighting conditions, improving operational efficiency in record verification.
- Collaborated with backend teams to integrate CV features into the main LIMS system, streamlining end-to-end workflow.

Stockerbd – Stock Market Tracker - Backend / ML 2024-2025

Project App: [Playstore](#) 

- Engineered a real-time backend architecture using Python and Redis, reducing app response time by 80% during peak traffic.
- Developed ML-based prediction modules and automated data pipelines for stock tracking, increasing user engagement through faster and reliable updates.
- Managed full deployment and monitoring, ensuring continuous service availability for thousands of concurrent users.

AI-Based Tour Guide – Research Project / Thesis - Full Stack 2024-2025

Project Link: [Live Project](#) 

- Led development of an AI-powered virtual tour guide using Python, NLP, and CV techniques to identify monuments and deliver detailed, context-aware information.
- Launched web beta version, receiving initial user testing feedback and iteratively improving recommendation accuracy by 20%.
- Coordinated with a teammate to implement interactive features, demonstrating end-to-end project ownership from research to deployment.

Technologies

Languages: Python, Flutter, Java

Technologies: MySQL, Postgres, Git, Docker, Jupyter Notebook, Redis, ChromaDB, Ollama

Framework and Libraries: Django, Flask, Numpy, Pandas, Plotly, Pytorch, Scikit Learn, TensorFlow, Keras, Streamlit, LangChain, pdfplumber

Other: RAG (Retrieval-Augmented Generation), Large Language Models (LLMs), Nomic Embeddings, OCR (Tesseract), Conversational AI, Data structures and algorithms, Object-Oriented Programming (OOP), Design Patterns, Software Development Life Cycle (SDLC), Continuous Integration/Continuous Deployment (CI/CD), Problem Solving, Debugging

Education

Daffodil International University

Sept 2017 – Feb 2023

BSc in Computer Science and Engineering

- CGPA: 3.70/4.0